CSE 592: Foundations of Artificial Intelligence
Project Milestone Report

Due Thursday, Nov 12th, 2026, 11:59 pm

For the project milestone, report on your group's progress so far. Submit a joint group report with up to 3 pages on Gradescope, covering the content we describe below. You can use additional space for references. You do not strictly need to fill 3 pages of content – your report will be graded only on whether it contains what we ask for below, and whether it is clearly written.

1. Project Title

Give a tentative title for your project, preferably short and specific, as if it was a scientific paper. You can keep the title from your proposal, or change it if you like.

2. Group

List again the name, UMID and uniqname of each member, same as in the proposal. If there were any changes to the group, note them here.

3. Problem motivation and description

Concretely describe the context around your problem of choice, either in AI or any other domain that you plan to approach using AI methods, as well as the problem itself. This description should now be ideally more concrete and specific than in the proposal. This section should cover:

- **Motivation.** Convey why your problem is important or interesting, perhaps with open questions you plan to investigate, or with examples of potential or existing applications.
- **Concrete problem statement.** State a specific, concrete problem, including what are the inputs to what you are building and what are its outputs. By now, you should ideally be able to give concrete examples of your problem. For instance, if you were building an image classifier, you could give examples of images in your domain and their labels. If you are building an LLM-based system, you can give concise but specific examples of the kinds of prompts and responses you expect your system to have.

4. Related Work

By now, you should be more familiar with papers that are related to what you are proposing. Describe them here, focusing on most recent work. For each paper you cite, both describe its main idea and how what you are proposing is different or complementary. This difference could be in the particular problem you are tackling, your method, analysis, or a combination of these.

5. Method

Concretely describe your proposed approach for solving the problem you are tackling. Make sure to define technical terms besides the basic vocabulary we saw in class — anything that is specific to your problem that a general AI audience would not be familiar with.

5. Preliminary Evaluation

Describe any preliminary results you have – perhaps for a simple baseline approach, and perhaps on small scale. The specifics of what makes sense to report depend on your project, but typically this will include (1) a description of a dataset, benchmark or task you are using for evaluation, (2) evaluation metrics, (3) a qualitative description of one or two examples that help a reader understand the system's current behavior (perhaps a success and a failure case, for instance).

6. Completion plan

Describe what next steps you hope to complete for the final report. It is OK if your scope changes based on your preliminary experiments (e.g., perhaps you realized your original approach will be difficult to scale and you now want to try something simpler).

Appendix: free and low-cost resources for your project

Below, we again give a list of free or low-cost resources you might find helpful for your project (primarily LLM APIs and GPU compute). It is not required that you use these resources. If you do, keep the *Warning* at the bottom in mind to avoid unexpected costs.

We also received a compute allocation in the U-M Great Lakes cluster for the final project. This can be used to run jobs using NVIDIA A100 and RTX6000 GPUs. This is a Slurm-managed cluster, where you run jobs primarily in batch mode, by writing a script that will be queued and run once resources are available. Instructions for Great Lakes are appended in this same PDF below – feel free to ask on Ed if you have questions, or during any of the GSI office hours.

Free notebooks and GPUs

- **Google Colab** — free GPU access on online Jupyter-style notebooks: https://colab.research.google.com
- **Kaggle Notebooks** — about 30 free GPU hours per week on P100 or T4 GPUs: https://www.kaggle.com/code
- **Hugging Face** — models, datasets, and Spaces (the free ZeroGPU tier gives short bursts on high-end hardware). https://huggingface.co
- **Lightning AI Studios** — a persistent VS Code-like workspace with 80 free hours on higher-end GPUs on sign-up: https://lightning.ai

Language model APIs

- **Google AI Studio / Gemini API** — offers a permanent free tier on the Gemini Flash models. https://aistudio.google.com/apikey.
- **Groq, Cerebras, OpenRouter** — free or near-free tiers for serving open-weight models.
- **Tinker** — flexible API for LLM training and inference using open-weight models, with fairly low cost for smaller models (e.g., Qwen3-8B or GPT-OSS-20B): https://tinker-docs.thinkingmachines.ai/tinker/models/

Student credits

- **GitHub Student Developer Pack** — verify with your `@umich.edu` address for GitHub Pro, Copilot, about $100 of Azure credit (where you can get GPUs or use various language model APIs). https://education.github.com/pack
- **New-account cloud provider trials** — Credits on Google Cloud ($300), Azure ($200), and AWS free tier.

Warning

Be *extremely careful* with setting spending limits on any service that might ask for a credit card. We highly recommend you estimate the cost of your experiments with very small pilots first. For instance, if your evaluation set has 1000 examples, you should make sure to run on 5-10 first and wait for the cost to show up before running a full evaluation. If you plan to train a model, train it on 1% of the dataset you plan to use first to get a sense of how long it will take or how much it will cost (e.g., if using Tinker), and budget accordingly. If you use language model APIs, we highly recommend you log and inspect the exact inputs and outputs to the model and make sure those are exactly what you intended them to be (e.g., inputs are not being accumulated or growing over time in a way you did not intend).

Information about Great Lakes Cluster

Zhengjie Xu

1 Introduction

We provide GPU access on Great Lakes cluster for both courses. If you do not have an ARC login please request one:

```
https://its.umich.edu/advanced-research-computing/high-performance-computing/great-lakes/getting-started
```

List Prof. Gabriel Poesia as your advisor and in the description say it is for the CSE592 class. If you already have a login, you do not need to request a new one. Once you have a login, to access the cluster:

```
ssh <uniqname>@greatlakes.arc-ts.umich.edu
```

Make sure you are on university Wi-Fi or on University of Michigan VPN. You can install VPN here:

```
https://its.umich.edu/enterprise/wifi-networks/vpn/getting-started
```

2 Storage

We have a directory with 10TiB of storage space:

```
/scratch/cse592f26_class_root/cse592f26_class
```

Within the directory, every student has a folder (name = uniqname) that only they can access.

3 Execution

3.1 Submit a job process

To run a program, submit a script as follows:

```
sbatch <shell script name>.sh
```

Here is a sample scritp:

```
#!/bin/bash
# The interpreter used to execute the script
#\#SBATCH" directives that convey submission options:
#SBATCH --job-name=example_job
#SBATCH --mail-user=uniqname@umich.edu
#SBATCH --mail-type=BEGIN,END
#SBATCH --cpus-per-task=8
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=1
#SBATCH --mem-per-cpu=1000m
#SBATCH --time=01:00:00
#SBATCH --account=account cse592f26_class
#SBATCH --partition=gpu_mig40,gpu,spgpu
#SBATCH -output=example_job.log
echo \hello world"
python3 yourcode.py
```

Activate the conda environment before submitting the script (or add `conda activate <your env name>` to the script). The output file will have the information your code prints.

3.2 Request an interactive job

It is also possible to request an interactive job (recommended for debugging):

```
salloc --account cse592f26_class
--partition gpu_mig40,gpu,spgpu --nodes 1 --ntasks 1
--cpus-per-task 1 --gpus 1 --mem 1G --time 00:00:10
```

You can see the status of your jobs using:

```
squeue -u <username>
```

You can see that status of all jobs on the class partition using:

```
squeue -A cse592f26_class
```

3.3 Limitation and rate

There is a limit of 2 jobs running at the same time and 1 GPU in use. Maximum duration is 8h. You have $60.91 in credits for the semester. Details of the cluster rates:

```
https://its.umich.edu/advanced-research-computing/high-performance-computing/great-lakes/rates
```

Cluster configuration (what sorts of GPUs it has):

```
https://documentation.its.umich.edu/node/4976
```

To get whichever gpu is free first of these options:

```
#SBATCH -p gpu_mig40,gpu,spgpu
```

How to check the cost of a job:

```
my_job_estimate -help
```

Acknowledgment

The content of this document is mostly based on the instruction document created by Anna Kay, Stanley Hsieh for EECS 542 in Fall 2024, 2025.